MILITARY SPECIFICATION

STRONTIUM PEROXIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers strontium peroxide for use in pyrotechnic mixtures.

1.2 Classification. Strontium peroxide shall be of the following grades, as specified (see 6.2):

Grade B
Grade C

1.2.1 Relationships of the grades of this specification to those of the previous revision are indicated in 6.5.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal
L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
RR-S-366 - Sieve, Test

Military
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-D-6054 - Drums, Metal-Shipping and Storage

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STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS
D 1193 - Reagent Water
E 300 - Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for referral from libraries. They are also distributed among technical groups and using Federal agencies.

DEPARTMENT OF TRANSPORTATION
49 CFR 170-189 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles
49 CFR 173.154 - Flammable Solids and Oxidizing Materials Not Specifically Provided For

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

3. REQUIREMENTS

3.1 Material. The strontium peroxide shall be a free flowing white powder.

3.1.1 Grade B strontium peroxide shall be manufactured by the wet precipitation method and shall consist essentially of anhydrous strontium peroxide.

3.1.2 Grade C strontium peroxide shall be manufactured by the wet precipitation method and shall consist of anhydrous strontium peroxide of a definite hygroscopicity requirement to control the stability of the material.

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3.2 Chemical and physical requirements. The chemical and physical requirements shall be in accordance with table I.

Table I. Chemical and physical requirements

Requirements	Grade B	Grade C
Available oxygen, min., percent	12.3	12.3
Moisture, max., percent	----	0.1
Hygroscopicity, max., percent	----	0.25
Apparent density, min., gm/ml	1.30-2.00	0.85
Barium, max., percent	1.0	0.45
Calcium, max., percent	0.3	0.15
Magnesium, max., percent	0.05	0.02
Sodium, max., percent	0.1	0.05
Iron, max., percent	0.09	0.03
Aluminum, max., percent	0.1	0.05
Nitrate, calculated as $Sr(NO_3)_2$, max., percent	0.1	0.05
Chloride, calculated as $SrCl_2$, max., percent	0.1	0.10
Total impurities other than SrO, max., percent	----	----

3.3 Granulation. Unless otherwise specified in the contract or order, each grade of strontium peroxide shall be furnished in accordance with the following specification.

Table II. Granulation U. S. Standard sieve (percent)

	Grade B	Grade C
Through No. 80	---	100
Through No. 100, min.	98	98
Through No. 140, min.	80	80

3.4 Workmanship. Strontium peroxide shall be free from grit, dirt, foreign matter or other injurious material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

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EXAMINE	DEFECT
Packing	Not as specified. Packing material not as specified. Closures not accomplished by specified methods or materials.
Marking	Not as specified. Marking wrong color. Marking incorrect size. Marking illegible, incorrect, incomplete, omitted or not in accordance with contract requirements.

4.2.3.2 Examination of end item. An examination of the end item shall be performed to determine the condition and appearance of the drummed product. The sample unit shall be one filled drum before closure. The inspection level shall be S-2 and the AQL 4.0 defects per 100 units.

EXAMINE	DEFECT
Appearance	Not material specified. Presence of foreign material. Caked or presence of lumps.
Loss of material	Evidence of loss or sifting of product from sealed drum.

4.2.3.3 Examination for contents of drum. The average net weight of contents of the drums sampled shall not be less than the amount specified in the contract or purchase order. The inspection level shall be S-2.

4.2.3.4 For test. Each of the two samples shall be subjected to all the tests specified in 4.3. If any sample is found to be not in conformance with the requirements of this specification, the lot represented shall be rejected. Where the material is produced by a continuous-run process, a rejected lot shall not be resubmitted for inspection. In case, however, material in the lot is identifiable by batch number, the manufacturer shall have the option or having partial or complete analysis of the batches of a rejected lot made at no expense to the Government. The manufacturer may then remove or replace defective portions of the lot and resubmit the lot for acceptance, provided the markings on the containers are such that complete removal of the defective portions of the lot can be made to the satisfaction of the inspector.

4.3 Test methods.

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4.3.1 General. Distilled water, in accordance with ASTM D 1193 Type III and analytical grade reagents, shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents used in the test, and corrections shall be applied when necessary.

4.3.1.1 Inspection Equipment. The government reserves the right to inspect contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in Section 4. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpoces.

4.3.2 Available oxygen.

4.3.2.1 Reagent. Acid reagent. Add 100 ml of perchloric acid, 30 ml of phosphoric acid, and 0.1 g of manganese chloride ($MnCl_2 \cdot 4H_2O$) to water, and dilute to 1 liter.

4.3.2.2 Procedure. Weigh 0.35 to 0.40 g of the sample on a tared watch glass. To a 300-ml tall form beaker add 70 ml of acid reagent. From a buret, run into the beaker about 3/4 of the amount of 0.1N potassium permanganate solution needed for the titration. Stir the solution in the beaker mechanically and carefully add the sample. Wash down the watch glass and sides of the beaker with a fine stream of acid reagent. Cover the beaker with a watch glass containing a hole for the tip of the buret and continue stirring. After solution of the sample is complete, titrate with the 0.1N potassium permanganate solution to a pink color that persists for 1 minute. Calculate the percent available oxygen as follows:

$$\text{Percent available oxygen} = \frac{0.8VN}{W}$$

where: V = ml of potassium permanganate solution.
N = normality of potassium permanganate solution.
W = g of sample.

NOTE: The above directions must be rigidly followed. The sample should be brought completely in solution as quickly as possible and then titrated immediately. Any delay in the titration will cause low results, regardless of stabilizer or lowering of the temperature. The solutions should not be much above room temperature (too high a temperature causes decomposition), nor below room temperature (too low a temperature retards dissolution of the sample).

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4.3.3 Moisture. Transfer a 3-g sample to a tared weighing dish (approximately 6 cm in diameter and 2.5 cm deep) and weigh. Dry at 130° ± 3°C for 3 hours, cool in a desiccator, and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\ A}{W}$$

where: A = g loss in weight.
W = g of sample.

Save the dried samples for the hygroscopicity determination (see 4.3.4).

4.3.4 Hygroscopicity. Place the dried samples from the moisture determination into a desiccator, 10-liter capacity, containing 1 liter of a sulfuric acid solution (30.1 ± 0.2% sulfuric acid by weight) for producing 75% relative humidity at 25°C. Allow the dishes to remain in the desiccator for 48 hours and weigh. Place the dishes back into the desiccator for 24 hours and weigh again. Repeat for 24-hour intervals until constant weight is obtained. Usually the weight will not change significantly after the initial 48-hour period. Calculate the percent hygroscopicity as follows:

$$\text{Percent hygroscopicity} = \frac{100\ A}{W}$$

where: A = g gain in weight.
W = g of sample.

4.3.5 Apparent density. Weigh a 50-ml graduated cylinder (ground-glass top) into which has been inserted a rubber stopper. Transfer about a 30-g sample to the cylinder by use of a powder funnel. With the stopper inserted, drop the cylinder vertically 30 times from a height of 2.0 inches onto a hard leather pad (or book), regulating the height of fall by inserting the cylinder through a support ring of suitable size. Avoid warming the cylinder by contact with the palms of the hands. Tap the sides of the cylinder gently once with a straight edge to level the height, read the volume occupied, and weigh. Calculate the apparent density as follows:

$$\text{Apparent density} = \frac{W}{V}$$

where: W = g of sample.
V = ml occupied.

4.3.6 Barium, calcium, magnesium, and sodium (see 6.2).

4.3.6.1 Apparatus. Atomic absorption spectrophotometer.

4.3.6.2 Reagents.

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4.3.6.2.1 Standard barium solution (1 ml = 1.0 mg Ba). Dissolve 1.7790 g of barium chloride ($BaCl_2 \cdot 2H_2O$) in water and dilute to 1 liter in a volumetric flask.

4.3.6.2.2 Standard calcium solution (1 ml = 0.5 mg Ca). Transfer 1.2490 g of calcium carbonate to a 400-ml beaker and add 100 ml of water. Add 5 ml of hydrochloric acid (slowly), heat to boiling, boil for 1 or 2 minutes, cool, and dilute to 1 liter in a volumetric flask.

4.3.6.2.3 Standard magnesium solution (1 ml = 0.05 mg Mg). Transfer 0.0500 g of magnesium ribbon to a 250-ml beaker and add 50 ml of water. Add 5 ml of hydrochloric acid (slowly), heat to boiling, boil for 1 or 2 minutes, cool, and dilute to 1 liter in a volumetric flask.

4.3.6.2.4 Standard sodium solution (1 ml = 0.05 mg Na). Dissolve 0.1271 g of sodium chloride in water and dilute to 1 liter in a volumetric flask.

4.3.6.3 Procedure. Transfer a 1-g sample to a 250-ml beaker, add 30 ml of nitric acid (1 to 9), warm to dissolve, and then boil down to a volume of about 10 ml. Cool and dilute to 100 ml in a volumetric flask. Dilute appropriate aliquots of the standard solutions of barium, calcium, magnesium, and sodium to 100 ml in volumetric flasks and measure the absorbance of the standards and the samples with the atomic absorption spectrophotometer in accordance with the manufacturer's recommendations. Calculate the mg of barium, calcium, magnesium, and sodium in the sample (per 100 ml) and calculate the percent of barium, calcium, magnesium, as follows:

$$\text{Percent barium, calcium, magnesium, and sodium} = \frac{M}{10W}$$

where: M = mg of barium, calcium, magnesium, and sodium (per 100 ml).
W = g of sample.

4.3.7 Iron (see 6.4).

4.3.7.1 Reagents.

4.3.7.1.1 Standard iron solution (1 ml = 0.1 mg Fe). Transfer 0.1000 g of pure iron to a 250-ml beaker, add 50 ml of hydrochloric acid and warm until dissolved. Add 3 ml of hydrogen peroxide (30%), boil 10 minutes, cool, and dilute to 1 liter in a volumetric flask.

4.3.7.1.2 Hydroxylamine hydrochloride solution (5%). Dissolve 25 g of hydroxylamine hydrochloride in water and dilute to 500 ml.

4.3.7.1.3 Ortho-phenanthroline solution (0.2%). Dissolve 1 g of ortho-phenanthroline in warm water, cool and dilute to 500 ml.

4.3.7.1.4 Sodium acetate solution (50%). Dissolve 250 g of sodium acetate ($CH_3COONa \cdot 3H_3O$) in water and dilute to 500 ml.

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4.3.7.2 Preparation of calibration curve. Transfer 0.5, 1.0, 2.0, 2.5, and 3.0 ml portions of standard iron solution to 100-ml volumetric flasks. Carry along a reagent blank. Add 50 ml of water and 5 ml of hydroxylamine hydrochloride solution (5%) and allow to stand 15 minutes. Add 5 ml of ortho-phenanthroline solution (0.2%) and 10 ml of sodium acetate solution (50%) and dilute to the mark. Allow to stand 30 minutes or more and measure the absorbance at 500 nm with a spectrophotometer or colorimeter that has been set to 100% transmittance with the reagent blank. Plot mg of iron against absorbance.

4.3.7.3 Procedure. Weigh a 1-g sample into a 400 ml beaker, add 50 ml of water and 5 ml of hydrochloric acid, and evaporate to incipient dryness on the hot plate at low heat. Carry along a reagent blank. Add 50 ml of water and warm to dissolve the salts. If the solution is not clear, filter through a Whatman No. 42 filter paper, and wash with water. Wash into a 100-ml volumetric flask (for Grade A strontium peroxide, dilute to the mark, pipet a 25-ml aliquot into a second 100-ml volumetric flask, and add 35 ml of water). Add 5 ml of hydroxylamine hydrochloride solution (5%) to the solution in the 100-ml volumetric flask and allow to stand 15 minutes. Add 5 ml of ortho-phenanthroline solution (0.2%) and 10 ml of sodium acetate solution (50%) and dilute to the mark. Allow to stand 30 minutes or more and measure the absorbance at 500 nm with a spectrophotometer or colorimeter that has been set to 100% transmittance with the reagent blank. Convert the reading to mg of iron by referring to the calibration curve. Calculate the percent iron as follows:

$$\text{Percent iron} = \frac{M}{10W}$$

where: M = mg of iron as read from calibration curve.
W = g of sample on which color was developed.

4.3.8 Aluminum (see 6.4). Weigh a 5-g sample into a 250-ml beaker, add 100 ml of water and 40 ml of hydrochloric acid, and heat on the hot plate until dissolved. Filter through a Whatman No. 42 filter paper into a 400-ml beaker and wash with water. Discard the precipitate. Add 4 g of ammonium chloride to the filtrate and boil for 5 minutes to destroy the peroxide. Add ammonium hydroxide until the solution is just alkaline to methyl red, boil for a minute, filter through a Whatman No. 40 filter paper, and wash with hot 1% ammonium chloride solution that has been made ammoniacal to methyl red. Dissolve the precipitate from the filter paper into the original beaker with about 20 ml of hot hydrochloric acid (1 to 1) and wash with water. Reprecipitate with ammonium hydroxide, filter, and wash with the hot 1% ammonium chloride wash solution. Place the filter paper and precipitate into a tared platinum crucible, char, and burn off the filter paper at low heat, and ignite at red heat for a few minutes. Add 2 drops of sulfuric acid and 3 ml of hydrofluoric acid and evaporate to dryness. Ignite at red heat for 30 minutes, cool, and weigh the combined oxides of aluminum and iron. Calculate the percent aluminum as follows:

$$\text{Percent aluminum} = \frac{52.9(A - 0.0143BW)}{W}$$

where: A = g of precipitate.
B = percent iron (as determined in 4.3.7).
W = g of sample.

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4.3.9 Nitrate. Weigh a 10-g sample into a 500-ml Kjeldahl flask and add 10 ml of water and 60 ml of glacial acetic acid. Carry along a reagent blank. Warm in a beaker of water at about 60°C to dissolve. If necessary, add up to 25 ml more acetic acid to complete the solution. Add a few drops of phenolpthalein indicator solution (0.1%) and then cautiously add sodium hydroxide solution (50%) until the color changes to pink. Add 5 ml excess sodium hydroxide solution. Add approximately 1 g of paraffin wax and 1 g of Devarda's alloy and immediately connect to a trap and a condenser, the end of which dips just below the surface of 50 ml of 4% boric acid solution contained in a 250-ml Erlenmeyer flask. Warm the Kjeldahl flask at approximately 80°C for 2 hours, then heat to boiling and distill until about 100 ml has been collected. Add 2 drops of bromophenol blue indicator solution to the distillate and titrate to a yellow color with 0.01N sulfuric acid. Calculate the percent nitrate as strontium nitrate as follows:

$$\text{Percent nitrate calculated as strontium nitrate} = \frac{10.58(V - v)N}{W}$$

where: V = ml of sulfuric acid for titration of sample.
v = ml of sulfuric acid for titration of blank.
N = normality of sulfuric acid.
W = g of sample.

4.3.10 Chloride. Transfer a 2-g sample to a nickel crucible and fuse with 10 g of sodium carbonate. Allow to cool, and dissolve the melt by boiling with about 200 ml of water. Filter through a Whatman No. 40 filter paper, wash with hot water, and discard the residue. Transfer the filtrate to a 500-ml Erlenmeyer flask. Add nitric acid (1 to 1) until the solution is acid and add 5 ml in excess. Add about 5 ml of 10% ferric nitrate solution. Add an excess of 0.1N silver nitrate solution, and titrate with 0.05N ammonium thiocyanate solution to the appearance of a brownish coloration that does not disappear on shaking. Determine the silver nitrate titer of the thiocyanate solution by diluting 20.00 ml of silver nitrate to 100 ml and titrating with the thiocyanate solution using 3 to 5 ml of the indicator. The titer is equal to 20.00 divided by the ml of ammonium thiocyanate solution used. Calculate the percent chloride as strontium chloride as follows:

$$\text{Percent chloride calculated as strontium chloride} = \frac{7.93(V-vF)N}{W}$$

where: V = ml of silver nitrate solution.
v = ml of thiocyanate solution.
F = silver nitrate titer of thiocyanate solution.
N = normality of silver nitrate solution.
W = g of sample.

4.3.11 Granulation. Nest the U. S. Standard sieve conforming to specification RR-S-366, properly superimposed, on a bottom pan. Place a 100-g portion of sample on the upper sieve. Add 2 metal disks about the size and weight of nickel coins. Cover and shake for 5 minutes by means of a mechanical shaker geared to produce 300 plus or minus 15 gyrations and 150 plus or minus 10 taps of the striker per minute. Weigh the amounts retained or passing through the sieves and calculate to percent on the basis of the total sample recovered.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Strontium peroxide shall be packed in 15 gallon (150 lbs) steel drums conforming to MIL-D-6054. Steel drums shall be with a twist lock or bolted ring closure. The lid shall have a tubular rubber gasket. The drums shall be provided with a bag liner fabricated from nominal 6 ml thick polyethylene. Polyethylene bags shall be made from plastic film conforming to L-P-378, type 1, finish 1. All bag seams and closure shall be heat sealed. Three units of desiccant conforming to MIL-D-3464 shall then be put on top of the sealed polyethylene bag (to absorb any moisture which may be entrapped). A printed card identified with marking in accordance with MIL-STD-129 shall be inserted just before the drum is closed.

5.1.2 Level C. Strontium peroxide shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediate use. Containers shall comply with Department of Transportation Regulations (see Code Federal Regulations CFR 170-189) and regulations of carriers as applicable to the mode of transportation.

5.2 Marking. In addition to any marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Strontium peroxide is used primarily in tracer igniters and to a lesser extent in basic tracer formulations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Grade of strontium peroxide desired.

(c) Level of packing required (see 5.1).

(d) Unit quantity required.

6.3 The barium, calcium, magnesium, and sodium can also be determined spectrographically.

6.4 The iron and aluminum can also be determined spectrographically or by atomic absorption. However, in case of dispute the samples shall be analyzed in accordance with 4.3.7 and 4.3.8.

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6.5 The relationships of the grades of MIL-S-612B to the previous revision MIL-S-612A are as follows:

MIL-S-612B	MIL-S-612A
Grade A deleted	Grade A
Grade B	Grade B
Grade C	Grade C

Custodians:
Army - MU

Preparing activity:
Army - MU

Review activities
Army - MU, PA, TE, MD
Navy - OS

Project No. 6810-B005

DSC - GS

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DOCUMENT IDENTIFIER AND TITLE

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO (If "Yes", in what way?)

4. REMARKS

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